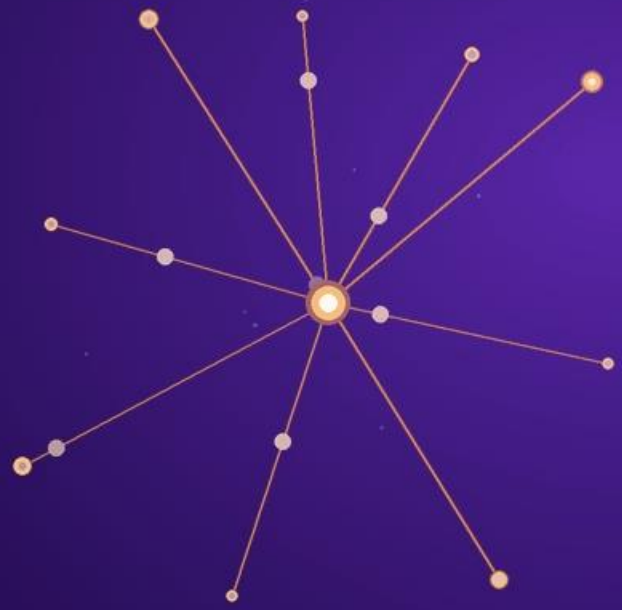
CORNELIS™

OCP GLOBAL SUMMIT 2026 – SAN JOSE

Changing Workloads Demand

ACF – Active Compute Fabric™



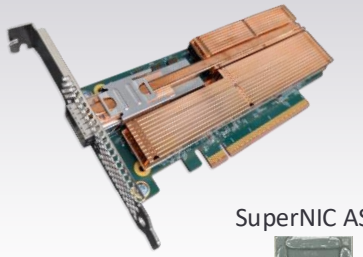
Jayraj Nair
Global Field CTO, **Cornelis**

CORNELIS™

Highest-Performance End-to-End Networking

SuperNIC's, Switches and Software

SuperNIC



SuperNIC ASIC



Complete Switch Portfolio

48-Port Switch



Switch ASIC



Up to 576-Port Director Switch



Optimized Open-Source Software

OPENFABRICS ALLIANCE

Lossless

Credit-based flow control. No drops, no pause-frame stalls.

Packet Spraying

Fine-grained adaptive routing on the best path, in real time.

Congestion-free

Telemetry and sender pacing stop incast before it spreads.

Resilient at Scale

Link-level retry and lane scaling to 500K+ endpoints.

Benchmarks: CN5000 vs. NVIDIA NDR 400G (ConnectX-7, Quantum-2), 2-socket AMD EPYC 9755, Intel MPI Benchmarks.

CORNELIS™

Maximizing Performance

Lower Latency and Higher Message Rate Drive Faster Applications

FABRIC ATTRIBUTES THAT MATTER FOR AI & HPC

UP TO

32%

Lower latency

Faster all-reduce for AI training and inference, and faster MPI collectives for HPC

UP TO

2.3x

Higher message rate

Handles the small, frequent messages of MoE, decode and tightly coupled HPC codes

UP TO

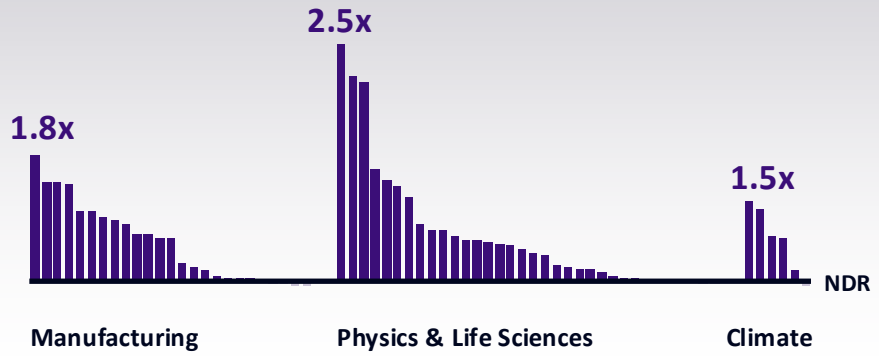
2.5x

Higher small-message bandwidth

More throughput for expert and KV-cache traffic, and for HPC halo exchanges

APPLICATION RESULT

HPC applications run faster than on NDR, by up to 2.5x.



Tests performed on 2 socket AMD EPYC 9755 128-Core Processor. Rocky Linux 9.5 (Blue Onyx). 5.14.0-503.33.1.el9_5.x86_64 kernel. 24x32GB, 768 GB total, Memory Speed: 6400 MT/s. Cornelis CN5000 - OPXS 12.1.x software release, Open MPI 5.0.8. NVIDIA NDR InfiniBand: Mellanox Technologies MT2910 Family [ConnectX-7]. MQM9700-NS2F Quantum 2 switch. 2M passive copper cables. UCX and Open MPI 4.1.9a1 as packaged in hpcx-v2.25.1. MLNX_OFED_LINUX-24.10-4.1.4.0. Application specific detail available upon request.

CORNELIS

CN5000 – Customer Success at Scale

Lawrence Livermore National Laboratory

900+ Servers

US National Nuclear Security

CTS-2, Lynx Cluster

#277 On Top500

91%** Scaling Efficiency

DELL intel

TACC
TEXAS ADVANCED COMPUTING CENTER

600+ Servers

Leading Academic Research

Stampede3 Cluster, Upgrade

Multi-Gen Collaboration

71%^ Higher WRF Perf at Scale

DELL intel



500+ Servers

German Science Research

AMD CPUs + Nvidia GPUs

Displaced InfiniBand

Nearing Production

AMD NEC

**Network scaling efficiency measured up to 900 nodes

^4 to 26 node WRF Conus2.5km Performance Scaling. vs. OPA100

CORNELIS™

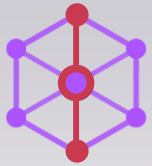
From Passive Pipe to Active Fabric



1ST GEN

Packet Fabric

Fixed-function ASICs. Forwards packets — nothing more.



2ND GEN

DPU / SmartNIC

Compute at the endpoints. Fixed function offload, but only at the edge.



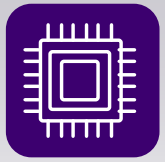
3RD GEN

Active Compute Fabric

A distributed compute fabric (DCF) adds cores on every NIC and switch. Your network becomes active; your idle silicon goes back to work.

CORNELIS

Changing Workloads – Fabric is the Glue



PREFILL

Compute-bound

Prompt processing saturates the matrix engines.



DECODE

Memory-bound

Token-by-token, limited by HBM bandwidth — not FLOPS.



MoE ROUTING

All-to-all

Tokens scatter to experts on other nodes - every layer.



AGENT LOOPS

Latency-critical

Multi-turn contexts of 100k+ tokens, reused prefixes.

Connected By The Fabric — Each pool scales independently.

Prefill Produces While Decode Consumes

PROMPT

PREFILL

Runs the whole prompt once, computing a Key + Value vector for every token.

KV CACHE

The attention memory of the sequence.

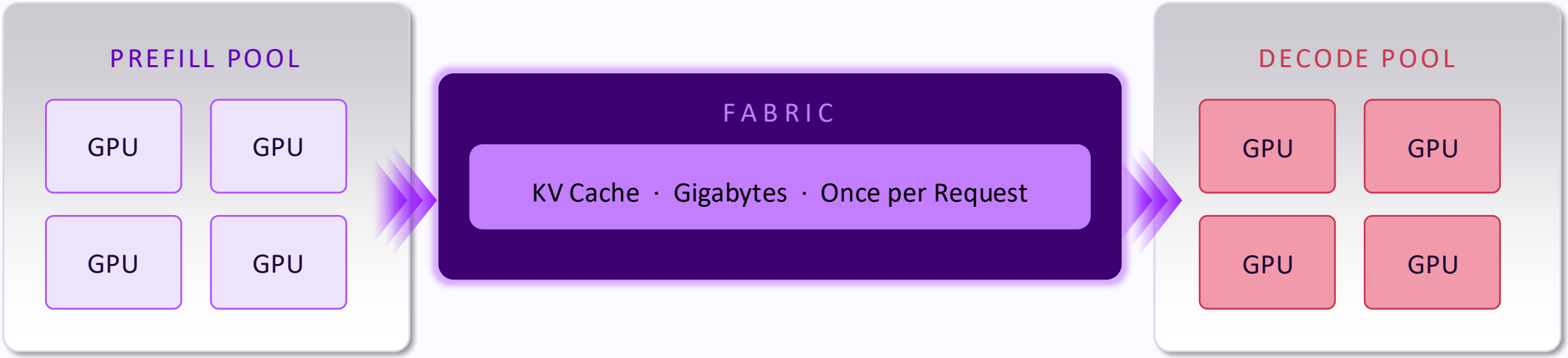
DECODE

Reads the KV cache on every step to generate the next token.

Two engines, one shared artifact between them —
the KV cache is the hinge of the whole computation.

KV Cache Hits the Wire

That transfer must finish before the first token. TTFT Is now A network number.



This isn't a bandwidth problem — it's a lack of intent in networking problem.
Exactly what Cornelis was created to solve.

CORNELIS™

Roadmap addresses changing workloads !

ROADMAP

CN5000 SHIPPING NOW

400G · PCIe 5.0 · Omni-Path

- Enterprise-class AI and HPC fabric
- Adaptive routing, advanced congestion and flow control

CN6000 SAMPLING NOW

800G · PCIe 6.0 · Multi-protocol

- Omni-Path, RoCEv2, Ultra Ethernet
- 1.6B msg/s, PQC root-of-trust
- Cornelis or 3rd-party Ethernet switches

CN7000 2027 TAPE OUT

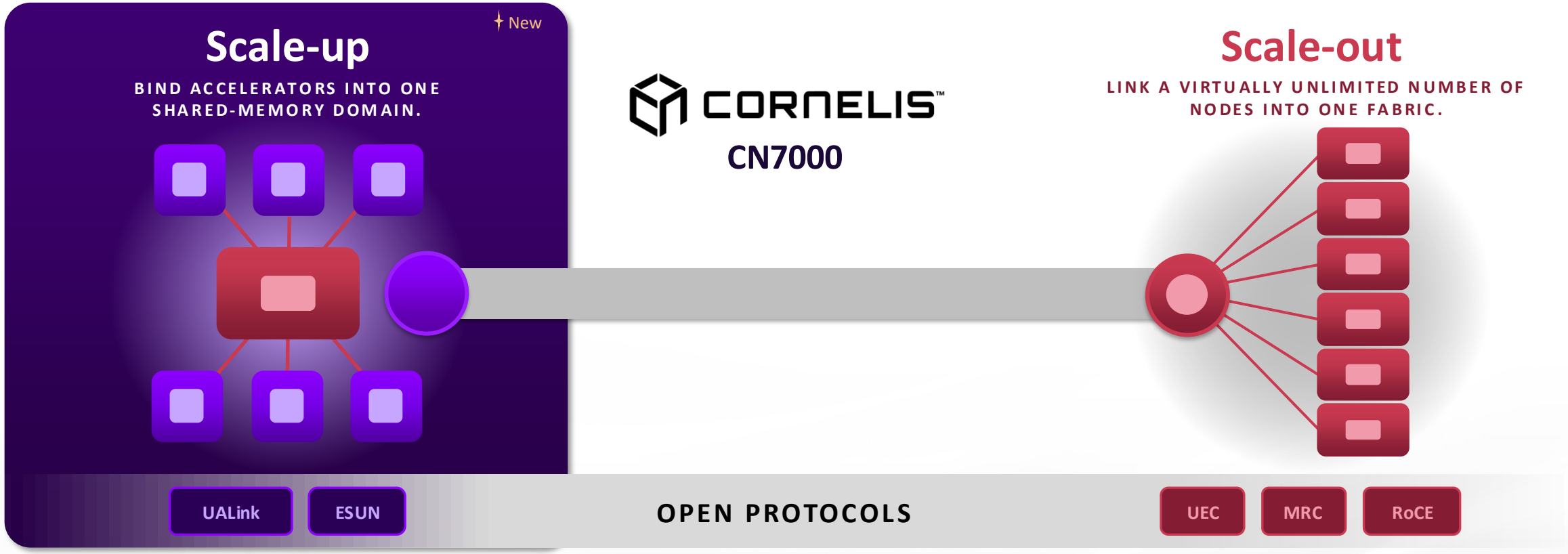
1.6T · PCIe 7.0 · 576 × 200G

- Scale-out: Ultra Ethernet + SONiC
- Scale-up: UALink and ESUN
- Distributed compute fabric

CN6000 expanded availability expected Q4 2026. CN6000 and CN7000 specifications are targets and subject to change.

CORNELIS™

AI Demands Scale-up and Scale-out



A distributed compute fabric adds cores at every NIC + Switch · Open protocols → choice · lower $/token.

CORNELIS™

CN7000 – Active Compute Fabric™

96	48 MB	72	8 + 16
CE cores	**on-switch SRAM**	**x 1.6T ports**	**cores per NIC**
per switch RV64GCV, VLEN=1024	line-rate accessible firmware-partitionable	per switch fabric standard Ethernet cabling	CE vector + SE scalar rich per-port telemetry

1.1 M
RISC-V cores @ 100K GPUs

~1.2 TB
distributed SRAM @ 100K GPUs

17 Pbps
aggregate injection BW @ 100K GPUs

Six DCF Acceleration Engines

KV Cache Acceleration	MoE Expert Dispatch	MPI / HPC Offload	PGAS / SHMEM Acceleration	In-Network Collectives	Checkpoint & Recovery
Accelerates LLM key-value transfers for faster Time-to-First-Token	Accelerates Mixture-of-Experts routing with in-network load balancing	Accelerates MPI progress via hardware tag matching.	Accelerates one-sided puts, gets, and atomic operations by eliminating host involvement	Accelerates AllReduce, Broadcast, and AllGather	In-fabric checkpoint, restore, and failure recovery

CORNELIS

No Lock-in: Open Standards Shape Buyer Choice

Ultra Ethernet Consortium

UALink™

OCP

ESUN

OPENFABRICS ALLIANCE

“Scaling Innovation for the AI Era”

Thank you

Jayraj Nair

Global Field CTO, Cornelis



CORNELIS™